

Question 1 (Norm increments in time and space.). If we want to keep the more general initial conditions u_0 and v_0 then we need to find an upper bound of the following form:

$$\int_{\mathbb{R}} |G(s, z - y) - G(t, x - y)| dy \leq C(T)(|s - t| + |z - x|) \quad \text{for } t \in [0, T] \text{ and } x, z \in \mathbb{R}.$$

To simplify this at first we can look at both

$$\int_{\mathbb{R}} |G(t, z - y) - G(t, x - y)| dy \tag{0.1} \quad \boxed{\text{E:HGspace}}$$

$$\int_{\mathbb{R}} |G(s, x - y) - G(t, x - y)| dy \tag{0.2} \quad \boxed{\text{E:HGtime}}$$

and in addition consider the Heat equation case,

$$G(t, x) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{x^2}{2t}\right).$$

Starting with (0.1), we first assume that $x < z$ and use the Mean Value Theorem to get

$$\int_{\mathbb{R}} |G(t, z - y) - G(t, x - y)| dy \leq |z - x| \int_{\mathbb{R}} \left| \frac{\partial G}{\partial x}(t, \xi(x, z) - y) \right| dy$$

where $\xi(x, z) \in (x, z)$ is a constant. Using Mathematica,

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g[x_] := (2 Pi t)^(-1/2) Exp[-(x^2)/(2t)]
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Integrate[Abs[D[g[x-y],x]],{y,-Infinity,Infinity}, Assumptions -> t>0
&& -Infinity < x < Infinity]
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we get that for any $x \in \mathbb{R}$ that

$$\int_{\mathbb{R}} \left| \frac{\partial G}{\partial x}(t, x - y) \right| dy = \frac{\sqrt{\frac{2}{\pi}}}{\sqrt{t}}$$

and hence

$$\int_{\mathbb{R}} |G(t, z - y) - G(t, x - y)| dy \leq \sqrt{\frac{2}{\pi}} \frac{1}{\sqrt{t}} |z - x|.$$

Similarly for (0.2), by assuming $t < s$ we get

$$\int_{\mathbb{R}} |G(s, x - y) - G(t, x - y)| dy \leq |s - t| \int_{\mathbb{R}} \left| \frac{\partial G}{\partial t}(\eta(t, s), x - y) \right| dy$$

where $\eta(t, s) \in (t, s)$ is a constant. And by using Mathematica,

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Integrate[Abs[D[g[x-y],t]],{y,-Infinity,Infinity}, Assumptions -> t>0
&& -Infinity < x < Infinity]
```

we get that for $t > 0$ and $x \in \mathbb{R}$ that

$$\int_{\mathbb{R}} \left| \frac{\partial G}{\partial t}(t, x - y) \right| dy = \sqrt{\frac{2}{e\pi}} \frac{1}{t}$$

and therefore

$$\int_{\mathbb{R}} |G(s, x - y) - G(t, x - y)| dy \leq \sqrt{\frac{2}{e\pi}} \frac{1}{\eta(t, s)} |s - t|.$$

Now we can see that

$$\begin{aligned} \int_{\mathbb{R}} |G(s, z - y) - G(t, x - y)| dy &\leq \int_{\mathbb{R}} |G(t, z - y) - G(t, x - y)| dy + \int_{\mathbb{R}} |G(s, z - y) - G(t, z - y)| dy \\ &\leq \sqrt{\frac{2}{\pi}} \frac{1}{\sqrt{t}} |z - x| + \sqrt{\frac{2}{e\pi}} \frac{1}{\eta(t, s)} |s - t|. \end{aligned}$$

The $1/\sqrt{t}$ and $1/\eta(t, s)$ seem to be an issue since they are unbounded as $t, s \rightarrow 0$.

Remark 0.1. We can tweak the upper bound for (0.2) a little to remove the $\eta(t, s)$. First note that

$$\begin{aligned} \left| \frac{\partial G}{\partial t}(t, x) \right| &= \left| -\frac{\exp\left(-\frac{x^2}{2t}\right)}{2\sqrt{2\pi}t^{3/2}} + \frac{\exp\left(-\frac{x^2}{2t}\right)x^2}{2\sqrt{2\pi}t^{5/2}} \right| \\ &\leq \frac{\exp\left(-\frac{x^2}{2t}\right)}{2\sqrt{2\pi}t^{3/2}} + \frac{\exp\left(-\frac{x^2}{2t}\right)x^2}{2\sqrt{2\pi}t^{5/2}} \end{aligned}$$

and notice that for $t < s$ we have that

$$\frac{\exp\left(-\frac{x^2}{2t}\right)}{2\sqrt{2\pi}t^{3/2}} + \frac{\exp\left(-\frac{x^2}{2t}\right)x^2}{2\sqrt{2\pi}t^{5/2}} \leq \frac{\exp\left(-\frac{x^2}{2s}\right)}{2\sqrt{2\pi}t^{3/2}} + \frac{\exp\left(-\frac{x^2}{2s}\right)x^2}{2\sqrt{2\pi}t^{5/2}}$$

and therefore

$$\begin{aligned} \int_{\mathbb{R}} \left| \frac{\partial G}{\partial t}(t, x - y) \right| dy &\leq \int_{\mathbb{R}} \frac{\exp\left(-\frac{(x-y)^2}{2s}\right)}{2\sqrt{2\pi}t^{3/2}} dy + \int_{\mathbb{R}} \frac{\exp\left(-\frac{(x-y)^2}{2s}\right)x^2}{2\sqrt{2\pi}t^{5/2}} dy \\ &= \frac{1}{2t} \sqrt{\frac{s}{t}} + \frac{s}{2t^2} \sqrt{\frac{s}{t}} \end{aligned}$$

which implies that

$$\int_{\mathbb{R}} |G(s, x - y) - G(t, x - y)| dy \leq \left(\frac{1}{2t} \sqrt{\frac{s}{t}} + \frac{s}{2t^2} \sqrt{\frac{s}{t}} \right) |s - t|.$$

However we still run into the same issue.

References

CHN19

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